No-Go Theorems for One-Step Lyapunov Potentials for the $3x + 1$ Map

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Abstract

A natural strategy for the $3x + 1$ (Collatz) problem is to exhibit a potential $\Phi(x) = \log_2 x + g(x)$ that is nonincreasing along the accelerated odd map $f(x) = (3x+1)/2^{\mathrm{v}_2(3x+1)}$, with a correction g computed from simple features of x. It is folklore that no such monotone invariant is known; we prove that for broad natural classes of corrections none can exist. The master mechanism is *shadowing an expanding rational cycle*: the cycle $-5 \mapsto -7 \mapsto -5$ has multiplier $3^2/2^3 > 1$, and its positive-integer shadows $x \equiv -5 \pmod{2^N}$ traverse its coordinate itinerary while gaining value, so any correction that cannot separate x from $f^2(x) = \frac{9}{8}x + \frac{5}{8}$ is refuted by two integers — the smallest certificate being $1275 \mapsto 1913 \mapsto 1435$. This excludes all corrections $g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{len}(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$, where τ is the trailing-ones count and the λ_i are suffix-pattern detectors of arbitrary depth (Theorem A). Iterating the shadow and applying Dirichlet approximation to $\log_2 \frac{9}{8}$ extends the exclusion to corrections using any fixed number of leading binary digits (Theorem B); a Mersenne burn argument excludes all bounded corrections of arbitrary dependence (Theorem C). The detectors are not ad hoc: they read a canonical structure, an exact ancestry tower $w_d(M) = \big(2^{M+2d} - 2^{2d+1} + 3^d\big)/3^d$ with $f^d(w_d(M)) = 2^M - 1$, supported on $M \equiv 1 \pmod{2 \cdot 3^{d-1}}$ with a carry-free periodic binary normal form (Theorem D). The surviving one-step class reads the quantized full logarithm, where further no-go progress becomes conjecture-adjacent.

1 Introduction

For odd $x \in \mathbb{Z}_{>0}$ let

$$f(x) = \frac{3x+1}{2^{\mathrm{v}_2(3x+1)}},$$

the accelerated (odd-to-odd) form of the Collatz map; the $3x + 1$ conjecture asserts that every orbit reaches 1. Among the strategies proposed for the problem, one of the most natural is the *potential-function* (Lyapunov) method: find $\Phi : 2\mathbb{Z} + 1 \to \mathbb{R}$ with $\Phi(f(x)) \le \Phi(x)$ for all odd $x > 1$, ideally of the coercive form $\Phi(x) = \log_2 x + g(x)$ with g a correction of lower order, so that monotonicity constrains orbit growth. Heuristics support the idea: a "random" odd step multiplies x by $3/2^e$ with e geometrically distributed of mean 2, so $\log_2 x$ decreases by $\log_2 \frac{4}{3}$ *on average* [5]. The difficulty, as surveys note, is that no correction making the decrease pointwise has ever been found [6]. To the best of our knowledge the literature contains no

*Repository: `github.com/docbgm2002/collatz-things`. Every numbered claim is machine-verified in exact integer or rational arithmetic; see Section 9.

theorem *excluding* a concrete class of such potentials; the absence is reported as an empirical fact. This paper proves exclusions, and locates the precise boundary at which the exclusion program must stop.

Throughout, $\tau(x) = \mathrm{v}_2(x+1)$ is the number of trailing `1`s in the binary expansion of odd x (the *fuel*: Lemma 2.3 shows an odd step converts one unit of fuel into a multiplication by almost exactly $\frac{3}{2}$); $\mathrm{len}(x)$ is the bit length; $\mathrm{LD}_j(x) = \lfloor x/2^{\mathrm{len}(x)-j} \rfloor$ is the top j bits; and $\lambda_1, \lambda_2, \ldots$ are the suffix-pattern detectors of Definition 2.5, which read the run length of a canonical hierarchy of periodic binary patterns (levels 1 and 2 being $(\texttt{10})^k\texttt{1001}$ and $(\texttt{111000})^k\texttt{01}$). Our main impossibility results:

Theorem A (master no-go; Theorem 3.2). For every $m, d \geq 0$ and every function g, the potential $\log_2 x + g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{len}(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$.

Theorem B (leading digits; Theorem 4.1). For every $m, j, d \geq 0$ and every g, the potential $\log_2 x + g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{LD}_j(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$.

Theorem C (bounded corrections; Theorem 5.1). For every bounded $G : 2\mathbb{Z}+1 \to \mathbb{R}$, of arbitrary dependence on x, the potential $\log_2 x + G(x)$ is not nonincreasing along f on odd $x > 1$.

Theorem A is proved by two integers per parameter choice: the map has the expanding rational cycle $-5 \mapsto -7 \mapsto -5$, of multiplier $3^2/2^3 = \frac{9}{8} > 1$ (expansion is exactly why it lives at negative integers), and its shadow at positive $x \equiv -5 \pmod{2^N}$ reproduces the cycle's coordinate itinerary while the value gains a factor $> \frac{9}{8}$ per period. Any correction blind to the difference between x and $f^2(x) = \frac{9}{8}x + \frac{5}{8}$ is refuted outright; the smallest certificate is $1275 \mapsto 1913 \mapsto 1435$. Theorem B iterates the shadow a times and chooses a, by Dirichlet approximation to the irrational $\log_2 \frac{9}{8}$, so that the accumulated gain sits next to an exact power of 2 and the leading digits return. Theorem C is a one-paragraph consequence of the Mersenne burn.

The detectors quantified over in Theorems A and B are given meaning by a structural result of independent interest:

Theorem D (Mersenne ancestry tower; Theorem 6.1). For $d \geq 1$ and $M \equiv 1 \pmod{2 \cdot 3^{d-1}}$, $M > 1$, the integer $w_d(M) = \big(2^{M+2d} - 2^{2d+1} + 3^d\big)/3^d$ is odd with $\tau = 1$, satisfies $3w_d(M) + 1 = 4\,w_{d-1}(M)$ and hence $f^d(w_d(M)) = 2^M - 1$ with every step of value ratio in $(\frac{3}{4}, 1)$; its binary expansion is exactly periodic (carry-free) of period $2 \cdot 3^{d-1}$; and $w_d(M) \equiv 1 - 2^{2d+1}3^{-d} \pmod{2^{M-1}}$, distinct levels separating 2-adically at bit $2\min(d, d') + 1$ exactly.

The tower is the complete uniform-e=2 ancestry of the Mersenne numbers — the slowest exact approach to maximal fuel — and λ_d detects membership in its level-d suffix pattern. The congruence domain is the order of 2 modulo 3^d, and the carry-free normal form is an instance of the lifting-the-exponent lemma.

Method and provenance

The constraint "Φ nonincreasing" restricted to any coordinate window is a difference-constraint system on the values of g; it is infeasible if and only if the coordinate graph of real orbit steps contains a directed cycle whose product of value ratios exceeds 1 (Section 7). Mining such cycles by Bellman–Ford over odd $x \leq 10^6$ produces startlingly short certificates — e.g. the pair $57 \mapsto 43$, $27 \mapsto 41$ already refutes all $g(x \bmod 16, \tau)$ — and inspection of the mined witnesses revealed that they shadow the -5 cycle, which led to the uniform theorems above. We record this provenance because the historical route of this project ran in the opposite direction: Theorems A–C were first proved for subclasses by designed families (a Mersenne "burn" that forces linear growth of g along a frozen coordinate line, against a "recharge" family that caps it), a method that yields an exact price list — each level of suffix detection buys a would-be potential exactly $\log_2 \frac{4}{3}$ of budget against an unbounded liability — but is dominated, as a source of impossibility statements, by the two-integer shadow certificates. The family analysis survives in the repository accompanying this paper and in the structural Theorem D.

The boundary

A coordinate defeats the shadow only by separating x from $f^{2a}(x)$ for every good Dirichlet approximation a: since the two agree on all low-order bits, fuel, suffix patterns, and (for suitable a) leading digits, the surviving one-step class reads the *quantized full logarithm* $\lfloor 2^j \log_2 x \rfloor$. There a closed certificate needs total gain in $(0, 2^{-j})$, which pure shadows cannot supply; and as $j \to \infty$ the class approaches arbitrary corrections, for which the existence of a monotone potential is conjecture-adjacent. Section 8 makes this precise.

Relation to the literature

The parity-vector structure underlying the heuristic goes back to Terras [12] and Everett [3]; density results culminate in [4] and, by different methods, in Tao's almost-all theorem [11]. Cycle exclusions rest on transcendence bounds [10, 9] and computation [2]. Backward (predecessor) structure of the Collatz graph is developed at the level of density and measure by Wirsching [13]; the tower of Theorem D is a distinguished explicit chain in that graph, and its closed form, congruence domain, and 2-adic freezing appear to be new. Forward identities of burn type, attached to the rational cycles at $-1, -5, -17$, appear in Andaloro [1] — the shadows of Section 3 are those identities viewed through the certificate lens. A search of the literature, including the annotated bibliographies [7, 8], located no prior impossibility theorems for concrete classes of one-step potentials; we would welcome references.

2 Preliminaries

Odd integers are written x; v_p is the p-adic valuation; $\tau(x) = \mathrm{v}_2(x+1)$; congruential inverses are taken in $(\mathbb{Z}/2^m\mathbb{Z})^\times$. Two standard facts are used.

Lemma 2.1 (LTE, special case)**.** *For even n, $\mathrm{v}_3(2^n - 1) = 1 + \mathrm{v}_3(n)$; in particular $\mathrm{v}_3\big(2^{2\cdot 3^{d-1}} - 1\big) = d$.*

Lemma 2.2. *2 is a primitive root modulo 3^d for every $d \geq 1$; hence $\mathrm{ord}_{3^d}(2) = \varphi(3^d) = 2 \cdot 3^{d-1}$.*

Proof. 2 generates $(\mathbb{Z}/9\mathbb{Z})^\times$ (its order divides 6 and is not $1, 2, 3$), and a primitive root mod 9 is a primitive root mod 3^d for all d. ☐

Lemma 2.3 (fuse)**.** *If $\tau(x) = t \geq 2$ then $\mathrm{v}_2(3x+1) = 1$, $f(x) = \frac{3x+1}{2}$, $\tau(f(x)) = t - 1$, and $f(x)/x = \frac{3}{2} + \frac{1}{2x}$.*

Proof. Write $x + 1 = 2^t u$, u odd. Then $3x + 1 = 2\,(3 \cdot 2^{t-1} u - 1)$ with odd bracket since $t \geq 2$, and $f(x) + 1 = 3 \cdot 2^{t-1} u$. ☐

Remark 2.4. The iterated form of Lemma 2.3 — that $2^t u - 1$ (u odd) iterates in t odd steps to $3^t u - 1$, the *burn* — is classical; it is recorded by Andaloro [1] (cf. the annotation in [8]).

Finally we define the detectors. For $d \geq 1$ write $\mathrm{ord}_d = 2 \cdot 3^{d-1}$ and, for $M \equiv 1 \pmod{\mathrm{ord}_d}$, $M > 1$,

$$w_d(M) = \frac{2^{M+2d} - 2^{2d+1} + 3^d}{3^d} = 1 + 2^{2d+1}\,\frac{2^{M-1} - 1}{3^d}, \tag{1}$$

an odd integer by Lemma 2.2 (Theorem 6.1 develops its structure; nothing beyond the formula and Lemma 2.6 below is needed before Section 6).

Definition 2.5. For $d \geq 1$ let $\beta_d(s)$ be the bit length of $T_s^{(d)} := w_d(1 + \mathrm{ord}_d\, s)$ and define the *level-d detector*

$$\lambda_d(x) = \max\bigl\{\, s \geq 1 : \ x \equiv T_s^{(d)} \ \bigl(\mathrm{mod}\ 2^{\,\beta_d(s)+1}\bigr)\bigr\} \ \in \ \mathbb{Z}_{\geq 1} \cup \{\perp\},$$

the maximum of the empty set being $\perp$.

Lemma 2.6. *Every template satisfies $T_s^{(d)} \equiv 9 \pmod{32}$ if $d = 1$ and $T_s^{(d)} \equiv 1 \pmod{32}$ if $d \geq 2$. Consequently $\lambda_d(y) = \perp$ for every d whenever $y \not\equiv 1, 9 \pmod{32}$.*

Proof. For $d \geq 2$, $2^{2d+1} \equiv 0 \pmod{32}$ in (1). For $d = 1$: M odd, $M \geq 3$, so $2^{M-1} \equiv 0 \pmod 4$, whence $(2^{M-1} - 1)/3 \equiv (-1) \cdot 3^{-1} \equiv 1 \pmod 4$ and $w_1(M) \equiv 1 + 8 \equiv 9 \pmod{32}$. Matching any template forces the mod-32 congruence, since every matching modulus $2^{\beta_d(s)+1}$ contains 32 ($\beta_d(s) \geq 4$). $\square$

3 The master theorem: shadowing the -5 cycle

The map has the rational cycle $-5 \mapsto -7 \mapsto -5$: $f(-5) = -7$, $f(-7) = -5$, with multiplier $3^2/2^3 = \frac{9}{8} > 1$. Expanding cycles live at negative integers (the cycle equation gives $x = c/(2^E - 3^K)$ with $2^E < 3^K$), and their positive shadows gain value while walking the cycle's coordinate itinerary.

Lemma 3.1 (shadow period)**.** *Let $N \geq 8$, w odd, and $x = 2^N w - 5$. Then*

$$\mathrm{v}_2(3x+1) = 1, \quad f(x) = 3 \cdot 2^{N-1} w - 7, \qquad \mathrm{v}_2(3f(x)+1) = 2, \quad f^2(x) = 9 \cdot 2^{N-3} w - 5,$$

so that $8\, f^2(x) = 9x + 5$ and $f^2(x)/x = \frac{9}{8} + \frac{5}{8x}$. Moreover $\tau(x) = \tau(f^2(x)) = 2$ and $\tau(f(x)) = 1$; $x \equiv f^2(x) \equiv -5$ and $f(x) \equiv -7 \pmod{2^{N-3}}$; and $\lambda_d(x) = \lambda_d(f(x)) = \lambda_d(f^2(x)) = \perp$ for every d.

Proof. Direct computation: $3x+1 = 2(3\cdot 2^{N-1}w - 7)$ with odd bracket; $3f(x)+1 = 4(9\cdot 2^{N-3}w - 5)$ with odd bracket; $8f^2(x) = 9 \cdot 2^N w - 40 = 9x + 5$. Fuel: $x + 1 = 4(2^{N-2}w - 1)$; $f(x) + 1 = 2 \cdot 3(2^{N-2}w - 1)$ with odd second factor; $f^2(x) + 1 = 4(9 \cdot 2^{N-5} w - 1)$ (odd bracket, $N \geq 6$). Residues are immediate. Detectors: $x \equiv f^2(x) \equiv 27$ and $f(x) \equiv 25 \pmod{32}$ (using $N - 3 \geq 5$), and $27, 25 \notin \{1, 9\}$; apply Lemma 2.6. $\square$

Theorem 3.2 (master no-go)**.** *Let $m, d \geq 0$ and let g be any real-valued function of $\bigl(x \bmod 2^m,\ \tau(x),\ \mathrm{len}(x),\ \lambda_1(x), \ldots, \lambda_d(x)\bigr)$. Then $\Phi(x) = \log_2 x + g(\cdots)$ is not nonincreasing along f on odd $x > 1$.*

Proof. Take $N \geq \max(m+3, 8)$, $k \geq 3$, and $w = 2^{k-1}+1$ in Lemma 3.1. Then additionally $\mathrm{len}(x) = \mathrm{len}(f^2(x)) = N+k$, since w places x in the lower part of its dyadic window: $9(2^k+2) < 2^{k+4}$. Hence x and $f^2(x)$ have identical coordinate vectors while $f^2(x) > x$, and summing the two monotonicity constraints at x and $f(x)$ gives $0 \leq -\log_2\big(f^2(x)/x\big) < -\log_2 \frac{9}{8} < 0$. □

The smallest member ($N = 8$, $k = 3$) is $1275 \mapsto 1913 \mapsto 1435$, with $8 \cdot 1435 = 9 \cdot 1275 + 5$ and ratio product $\frac{287}{255}$: two integers refute every g on the full coordinate algebra at $m \leq 5$.

Remark 3.3 (general principle). Nothing is special about -5: every expanding rational cycle of the map — every solution of the cycle equation with $2^E < 3^K$, e.g. the $K = 7$ cycle at -17 with multiplier $3^7/2^{11}$ — shadows at positive integers $\equiv$ (cycle member) mod 2^N into a closed coordinate itinerary of value gain $3^K/2^E > 1$, refuting every correction that is 2-adically local along it. Expanding negative cycles are certificate factories against local potentials.

4 Iterating the shadow: leading digits

A coordinate can defeat a single shadow period only by separating x from $f^2(x) = \frac{9}{8}x + \frac{5}{8}$; the leading digits do (the ratio $\frac{9}{8}$ shifts them). Iteration removes this refuge.

Theorem 4.1 (leading digits do not survive)**.** *For every $m, j, d \geq 0$ and every g, the potential $\log_2 x + g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{LD}_j(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$.*

Proof. For $x \equiv -5 \pmod{2^{3a+c}}$, $c \geq 5$, Lemma 3.1 applies a times (each period consumes three bits of shadow depth), the itinerary alternating residues $-5, -7$ with fuel $2, 1$ and every detector $\perp$ throughout, giving the exact closed form

$$f^{2a}(x) = \frac{9^a x + 5\,(9^a - 8^a)}{8^a} = \left(\tfrac{9}{8}\right)^a x\,(1+\delta), \qquad 0 < \delta < \tfrac{5}{x}.$$

Now $\beta = \log_2 \frac{9}{8}$ is irrational ($2^{\,p+3q} = 3^{\,2q}$ is impossible: v_3 of the sides differ), so by Dirichlet there is $a \geq 1$ with $\|a\beta\| < 2^{-j-3}$ (distance to the nearest integer P; note $a\beta > 0$ regardless of the side). Choose $N = 3a + \max(m, 5) + j + 8$ and a free odd multiplier w placing the fractional part of $\log_2 x$, $x = 2^N w - 5$, at distance $> 2^{-j-2}$ from every breakpoint of LD_j (possible since $\log_2(2^N w - 5) \approx N + \log_2 w$ and $\{\log_2 w\}$ is dense), with N large enough that $\log_2(1+\delta) < 2^{-j-3}$. Then $\log_2 f^{2a}(x) = \log_2 x + P \pm \|a\beta\| + \log_2(1+\delta)$, so LD_j closes; all other coordinates close by the itinerary; and telescoping monotonicity over the $2a$-step walk gives $0 \geq a\beta + \log_2(1+\delta) > 0$. □

A concrete instance ($j = 4$): $a = 53$ gives $53\beta = 9.0060\ldots$; the verifier realizes the full 106-step walk in exact integers at depth $N = 177$, with top-4 bits `1011` $\to$ `1011`, all coordinates frozen, and value gain $2^{9.006}$.

5 Bounded corrections

Theorem 5.1 (no bounded correction)**.** *Let $G : 2\mathbb{Z}+1 \to \mathbb{R}$ be any bounded function. Then $\log_2 x + G(x)$ is not nonincreasing along f on odd $x > 1$.*

Proof. Let $\Omega = \sup G - \inf G$. Telescoping monotonicity over the burn of $2^M - 1$ (Lemma 2.3 iterated: $M-1$ steps, each of ratio $> \frac{3}{2}$) forces $(M-1)\log_2 \frac{3}{2} \leq \Omega$, false for $M > 1 + \Omega/\log_2 \frac{3}{2}$. □

Theorems A–C partition the correction landscape: bounded corrections fail on the burn alone; unbounded corrections through residues, fuel, length, leading digits, and suffix detectors of every depth fail on the shadow. Section 8 describes what is left.

6 The Mersenne ancestry tower

We now develop the structure behind the detectors. Define $w_0(M) = 2^M - 1$ and $w_d(M)$ as in (1).

Theorem 6.1. *For $d \geq 1$ and $M \equiv 1 \pmod{\mathrm{ord}_d}$, $M > 1$:*

(a) (Domain.) 3^d *divides* $2^{M+2d} - 2^{2d+1} + 3^d$ *iff* $M \equiv 1 \pmod{\mathrm{ord}_d}$.

(b) (Form.) $w_d(M)$ *is an odd integer* > 1 *with* $\tau(w_d(M)) = 1$.

(c) (Descent.) $3\,w_d(M) + 1 = 4\,w_{d-1}(M)$*; hence* $\mathrm{v}_2(3w_d(M)+1) = 2$, $f(w_d(M)) = w_{d-1}(M)$, *and* $f^d(w_d(M)) = 2^M - 1$, *each step of value ratio* $\frac{3}{4} + \frac{1}{4w_d} \in (\frac{3}{4}, 1)$.

(d) (Normal form.) $B_d := (2^{\mathrm{ord}_d} - 1)/3^d$ *is an integer in* $(0, 2^{\mathrm{ord}_d})$ *by Lemma 2.1, and for* $M = 1 + \mathrm{ord}_d\, s$,

$$w_d(M) = 1 + 2^{2d+1} \cdot \big[\textit{the } \mathrm{ord}_d\textit{-bit block of } B_d\big]^s,$$

the bracket denoting s carry-free repetitions: an exactly periodic binary word of period $2 \cdot 3^{d-1}$.

(e) (Freeze and separation.) $w_d(M) \equiv \Lambda_d := 1 - 2^{2d+1}3^{-d} \pmod{2^{M-1}}$*; and for* $d \neq d'$, $\mathrm{v}_2(\Lambda_d - \Lambda_{d'}) = 2\min(d, d') + 1$.

Proof. (a) Modulo 3^d the numerator is $2^{2d}(2^M - 2)$, so divisibility is $2^{M-1} \equiv 1 \pmod{3^d}$; apply Lemma 2.2. (b) The second form in (1) is 1 plus an even number, and $w_d + 1 = 2(1 + 2^{2d}q)$ with $2^{2d}q$ even. (c) Clearing 3^{d-1}, the identity reduces to $3^d + 3^{d-1} = 4 \cdot 3^{d-1}$; w_{d-1} is odd, so $\mathrm{v}_2 = 2$ exactly; iterate. (d) With $M - 1 = \mathrm{ord}_d\, s$, $(2^{M-1} - 1)/3^d = B_d \sum_{i=0}^{s-1} 2^{\mathrm{ord}_d\, i}$, a base-$2^{\mathrm{ord}_d}$ number all of whose digits equal $B_d < 2^{\mathrm{ord}_d}$: no carries; the $+1$ and the shift by 2^{2d+1} cannot interact. (e) $3^d \cdot \frac{2^{M-1}-1}{3^d} \equiv -1 \pmod{2^{M-1}}$; and for $d > d'$, $\Lambda_d - \Lambda_{d'} = 2^{2d'+1}3^{-d}\big(3^{\,d-d'} - 2^{\,2(d-d')}\big)$ with odd bracket. □

Levels $0, 1, 2$ are the Mersennes, the family $x'_M = (2^{M+2} - 5)/3 = (\mathtt{10})^{(M-3)/2}\mathtt{1001}$ with $f(x'_M) = 2^M - 1$, and $(\mathtt{111000})^k\mathtt{01}$; the tower is the complete uniform-e=2 ancestry of maximal fuel, and every tower member lies on the residue class $8y + 1$. Composing the tower with the burn (Remark 2.4) yields, for each (d, M), an exactly known orbit segment of length $d + M + 1$ terminating on the base-3 repunit $(3^M - 1)/2$.

Lemma 6.2. *(1) $\lambda_d(T_s^{(d)}) = s$: the detector reads its own tower index, unboundedly. (2) For $d \neq d'$, with $\nu = 2\min(d, d') + 1$: for all $M > \nu + 1$ in the level-d progression, $\lambda_{d'}(w_d(M)) \leq \nu / \mathrm{ord}_{d'}$.*

Proof. (1) At index s the congruence is an equality that holds; at $s' > s$ the modulus exceeds both operands and templates strictly increase. (2) Suppose $w_d(M)$ matches $T_s^{(d')}$ with $M' = 1 + \mathrm{ord}_{d'}\, s$ and, toward a contradiction, $M' - 1 \geq \nu + 1$. The matching modulus refines $2^{\nu+1}$ (as $\beta_{d'}(s) \geq M'$), so Theorem 6.1(e) on both sides forces $\Lambda_d \equiv \Lambda_{d'} \pmod{2^{\nu+1}}$, contradicting $\mathrm{v}_2(\Lambda_d - \Lambda_{d'}) = \nu$. Hence $\mathrm{ord}_{d'}\, s \leq \nu$. (Small-index templates have not converged to $\Lambda_{d'}$; this affects which small values occur, not the bound.) □

Remark 6.3 (the family method and the price list). Historically, Theorems A–C were first obtained for subclasses by a family method that yields quantitative structure the shadow does not. The *burn* family $3^j 2^t - 1$ ascends at ratio $> \frac{3}{2}$ per step with frozen residue -1 and $\lambda \equiv \perp$, forcing g to grow at slope $\log_2 \frac{3}{2}$ along a coordinate line; the tower's e=2 chains cap the same values from above, each level costing one $\log_2 \frac{4}{3}$. The resulting price list — each detector level buys a would-be potential exactly $\log_2 \frac{4}{3}$ of budget against an unbounded liability — explains *why* suffix information fails, while the shadow certificates prove *that* it fails. The full family proofs are retained in the repository.

7 Certificates from raw orbits

The constraint system "Φ nonincreasing", restricted to any coordinate window, is a system of difference constraints $g(v) - g(u) \le -\log_2(f(x)/x)$ over observed steps with $u = \mathrm{coords}(x)$, $v = \mathrm{coords}(f(x))$; it is infeasible iff the coordinate graph contains a directed cycle whose product of value ratios exceeds 1, and any such cycle is verifiable in exact rational arithmetic. Mining by Bellman–Ford over odd $x \le 10^6$ yields short certificates in every window we tested: the pair $57 \mapsto 43$, $27 \mapsto 41$ closes a 2-cycle in $(x \bmod 16, \tau)$-space with product $\frac{1763}{1539} > 1$, refuting Theorem 3.2's class at $m \le 4$ in two lines; a 5-cycle through $41 \mapsto 31 = 2^5 - 1$ does the same with the level-1 detector added. Inspection of mined witnesses at larger moduli revealed the $-5, -7$ residue pattern of Section 3. This pipeline is logically independent of every lemma in this paper — it uses raw orbit steps only — and would have refuted the theorems had a valid correction existed on the tested windows.

8 The boundary

What survives Theorem 4.1 must see $\log_2 x$ to a precision finer than every $\|a\beta\|$ — integer and fractional parts jointly: the *quantized full logarithm*. There, a closed certificate needs total gain in $(0, 2^{-j})$, which pure shadows cannot supply ($\beta > 2^{-j}$ for $j \ge 3$); any certificate must mix gain and loss segments, a connectivity question in the coordinate graph, minable per window but with no uniform construction known.

Question 8.1. Is there a nonincreasing potential of the form $\log_2 x + g\big(\lfloor 2^j \log_2 x \rfloor,\ \tau(x),\ x \bmod 2^m\big)$ for some j, m? As $j \to \infty$ the class approaches arbitrary corrections, for which the existence of a monotone potential is conjecture-adjacent; the one-step no-go program has plausibly reached the point where further progress and the conjecture itself interact.

Question 8.2. Do multi-step potentials (functions nonincreasing along f^k for fixed k, or path-dependent accountings) admit analogous obstructions? The shadow supplies candidate constraints, since its k-step behaviour is exactly known; note that any k-step potential must in particular survive k dividing $2a$ along the iterated shadow.

We also note what the results do *not* say: nothing here bears on individual trajectories, densities of divergent orbits, or cycles. The theorems delimit a proof *method*, not the dynamics.

9 Computational verification

Every numbered claim is verified by scripts using exact integer and rational arithmetic (no floating point in any assertion): the shadow lemma over wide parameter ranges with detector evaluation at all nodes (`verify_shadow.py`); the iterated shadow, the concrete Dirichlet instance $(m, j, a) = (6, 4, 53)$ at depth $N = 177$, and the irrationality arithmetic (`verify_leading_digit.py`); the tower in all parts, including both directions of the domain criterion and the carry-free normal form (`verify_tower.py`); the family-method subclasses (`verify_no_local_potential.py`, `verify_nlp2.py`); and the independent certificate miner with exact cycle confirmation (`verify_nogo_certificat` `verify_fuel_fraction.py`). The scripts accompany the repository.

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